

P1.

$$V = \text{span} \{(1, 1, 0), (1, 0, 1)\}; x = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

$$v_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

Let the projection be $(c_1 v_1 + c_2 v_2)$

$$\begin{aligned} \therefore \text{proj}_V(x) &= c_1 v_1 + c_2 v_2 \\ &= (c_1 + c_2, c_1, c_2) \end{aligned}$$

The error must be perpendicular to both v_1, v_2

$$(x - \text{proj}_V x) \cdot v_1 = 0$$

$$\Rightarrow [c_1 + c_2 \quad c_1 \quad c_2] \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} = 0$$

$$\Rightarrow 2c_1 + c_2 = 0$$

$$\Rightarrow [1 - c_1 - c_2 \quad 2 - c_1 \quad 3 - c_2] \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} = 0$$

$$\Rightarrow 1 - c_1 - c_2 + 2 - c_1 = 0$$

$$\Rightarrow 2c_1 + c_2 = 3 \quad \text{--- (1)}$$

$$(x - \text{proj}_V x) \cdot v_2 = 0$$

$$\Rightarrow [1 - c_1 - c_2 \quad 2 - c_1 \quad 3 - c_2] \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = 0$$

$$\Rightarrow 1 - c_1 - c_2 + 3 - c_2 = 0$$

$$\Rightarrow c_1 + 2c_2 = 4 \quad \text{--- (2)}$$

Solving, $c_1 = \frac{2}{3}, \quad c_2 = \frac{5}{3}$

$$\therefore \text{proj}_U(n) = \left(\frac{7}{3}, \frac{2}{3}, \frac{5}{3} \right)$$

(Ans)

P2.

$$U = \text{span} \{ (1, 2, 1), (0, 1, -1) \}$$

$$\text{Let } w = (x, y, z) \in U^\perp$$

w must be $\perp$ to both spanning vectors

$$(x, y, z) \cdot (1, 2, 1) = 0$$

$$\Rightarrow x + 2y + z = 0 \quad \text{--- (1)}$$

$$(x, y, z) \cdot (0, 1, -1) = 0$$

$$\Rightarrow y - z = 0 \quad \text{--- (2)}$$

From (1) and (2),

$$y = z \quad ; \quad x = -3y = -3z$$

$$\text{Let } y = t, \quad w = (-3t, t, t) = t(-3, 1, 1)$$

$$U^\perp = \text{span} \{ (-3, 1, 1) \}$$

$$\text{Norm}(U^\perp) =$$

$$\text{Norm}(U^\perp) = \sqrt{11}$$

$\therefore$ One orthonormal basis of $U^\perp$ is

$$\left\{ \frac{1}{\sqrt{11}} (-3, 1, 1) \right\}$$

(Ans)

p3.

$$Q = \frac{1}{3} \begin{bmatrix} 2 & -1 & 2 \\ 2 & 2 & -1 \\ -1 & 2 & 2 \end{bmatrix}$$

$$\text{Let } M = \begin{bmatrix} 2 & -1 & 2 \\ 2 & 2 & -1 \\ -1 & 2 & 2 \end{bmatrix} = 3Q$$

$$M^T = \begin{bmatrix} 2 & 2 & -1 \\ -1 & 2 & 2 \\ 2 & -1 & 2 \end{bmatrix}$$

$$M^T M = \begin{bmatrix} 4+4+1 & -2-2+4 & -2+4-2 \\ -2+4-2 & 4+4+1 & -2+4-2 \\ 4-2-2 & -2-2+4 & 4+1+4 \end{bmatrix}$$

$$= \begin{bmatrix} 9 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 9 \end{bmatrix}$$

$$= 9I$$

$$Q^T Q = \frac{1}{9} M^T M = I$$

$\therefore Q$ is orthogonal

proved

p4.

$$A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

$$\det \begin{bmatrix} 1-\lambda & 1 \\ 1 & 1-\lambda \end{bmatrix} = 0$$

$$\Rightarrow (1-\lambda)^2 - 1 = 0$$

$$\Rightarrow \lambda^2 - 2\lambda = 0$$

$$\Rightarrow \lambda = 0, 2$$

For $\lambda = 0$,

$$Av = 0$$

$$\Rightarrow x + y = 0$$

$$\Rightarrow x = -y$$

$$\text{let } x = 1$$

$$\therefore v_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$\|v_1\| = \sqrt{2}$$

$$\therefore \textcircled{v_1} = q_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$\therefore P = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$$

Diagonal matrix,

$$\boxed{D} = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 2 \end{bmatrix}$$

$$\text{Ans. } A = P D P^T$$

$$= \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$$

$$= \frac{1}{2} \begin{bmatrix} 0 & 2 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$$

$$= \frac{1}{2} \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

verified

For $\lambda = 2$,

$$\begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0$$

$$\Rightarrow x - y = 0$$

$$\Rightarrow x = y$$

$$\text{let } x = 1$$

$$v_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\|v_2\| = \frac{1}{\sqrt{2}}$$

$$\therefore q_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

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$$S = \text{span} \{ (1, 1, 1) \}$$

$$x = (2, 1, 3)$$

$$\text{proj}_S(x) = \frac{x \cdot v}{v \cdot v} v$$

$$= \frac{6}{3} (1, 1, 1) =$$

$$(2, 2, 2)$$

The perpendicular component is,

$$x - \text{proj}_S(x) = (2, 1, 3) - (2, 2, 2) =$$

$$(0, -1, 1)$$

∴ The shortest distance

$$= \sqrt{0^2 + (-1)^2 + 1^2}$$

$$= \sqrt{2}$$

(Answer)